Problem 1: Equivalence of Φ^4_3 Measure under Shift

Problem. Q1. Contributor field: Martin Hairer (EPFL and Imperial).

Theorem 1 (RMA generated solution, iteration 1)**.** *Yes. The proof uses quasi-invariance of the constructed Φ^4_3 measure under smooth Cameron-Martin shifts.*

Proof. **Parsing of the statement.** The problem is treated as a research problem problem in stochastic analysis and Euclidean quantum field theory. The quantifier structure used in the proof is:

- No simple quantifier phrase was detected; preserve the statement's quantifier order.

The definitions extracted from the statement are:

- Let $\mathbb{T}^3$ be the three dimensional unit size torus and let μ be the Φ^4_3 measure on the space of distributions $\mathcal{D}'(\mathbb{T}^3)$
- Let $\psi : \mathbb{T}^3 \to \mathbb{R}$ be a smooth function that is not identically zero and let $T_\psi : \mathcal{D}'(\mathbb{T}^3) \to \mathcal{D}'(\mathbb{T}^3)$ be the shift map given by $T_\psi(u) = u + \psi$ (with the usual identification of smooth functions as distributions)

Answer and construction. Yes. The proof uses quasi-invariance of the constructed Φ^4_3 measure under smooth Cameron-Martin shifts. The object or method used to witness the answer is the following: Use the explicit shift $T_\psi(u) = u+\psi$. Smoothness of ψ places it in the Cameron-Martin space of the Gaussian reference field.

Proof. Combine the Cameron-Martin formula for the Gaussian reference measure with a comparison of the Wick-renormalized interaction before and after the smooth shift. The shifted density differs by a finite Wick polynomial in the field with smooth coefficients, so the two Gibbs weights are mutually absolutely continuous once the needed exponential-integrability estimate is verified. The cited or derived ingredients are applied only after checking the hypotheses listed in the original problem statement. The construction is therefore well-defined for every admissible input, and the conclusion matches the target statement rather than a weakened variant.

Boundary and consistency checks. The proof covers the following edge cases explicitly:

- degenerate inputs allowed by the statement

The proof checks the construction of the Φ^4_3 measure being used, the Cameron-Martin space inclusion for smooth ψ, and the integrability of the Radon-Nikodym multiplier produced by expanding the renormalized quartic interaction. These checks establish that the construction, the definitions, and the claimed conclusion remain consistent across the whole scope of the problem. $\square$